

Collection of EDA Tools for Circuit/Embedded Designing

Schematic editor & simulator	BSch3V schematic capture package by Sugiyodo Online	Free
	Cider	Free
	ExpressSch by ExpressPCB	Free
	FidoCadJ by Davide Bucci	Free
	KtechLab	Free
	Logic Gate Simulator by South Puget Sound Community College	Free
	Logically by Joshua Tynjala	Licensed
	Ngspice	Free
	Qucs (Quite Universal Circuit Simulator)	Free
	Simulide	Free
	Spice	Free
	sPlan	Licensed
	TinyCAD	Free
	Allegro PCB SI by Cadence	Licensed
Signal integrity	HyperLynx SI by Mentor Graphics	Licensed
	Quantum SI by SiSoft	Licensed
	Signal Integrity Studio by Teledyne LeCroy	Licensed
	Signity Power SI by Cadence	Licensed
	Zeni SI by Zeni EDA	Licensed
Schematic editor and simulator	5Spice by Richard Andresen	Licensed
	CircuitLab by CircuitLab, Inc. (online)	Licensed
	DoCircuits by Sparsha Learning Technologies Pvt Ltd (online and offline)	Licensed
	LTSlice IV by Mike Engelhardt, Linear Technology Corporation	Free
	Logisim by Dr Carl Burch, Hendrix College	Free
	NI MultiSIM by National Instruments	Licensed
	OpenSce	Free
	SmartSim by Ashley Newson (available for Raspberry Pi also)	Free
	Sprint-Layout	Licensed
	Webench Schematic Editor by Texas Instruments (online)	Free
PCB design and layout	ExpressPCB by ExpressPCB	Free
	FreePCB	Free
	PCB—Printed Circuit Board Editor	Free
	ZeniPCB by Stortini Mirko Bruno	Free
Design Suite	AutoTRAX Design Express (DEX) by DEX 2020 (AutoTRAX Software)	Licensed
	CADSTAR, CR-5000 and CR-8000 by Zuken	Licensed
	CADSTAR Express by Zuken	Free
	CST STUDIO SUITE by CST	Licensed
	DesignSpark PCB by RS Components and Allied Electronics	Free
	DipTrace by Novarm	Licensed
	EAGLE PCB Software by CadSoft	Licensed
	Fritzing (particularly for Processing and Arduino communities)	Free
	gEDA	Free
	KiCad EDA Software Suite	Free
	NI Ultiboard by National Instruments	Licensed
	OrCAD by Cadence	Licensed
	Oscad	Free
	Osmond PCB by J.C. Chavez (for Mac)	Licensed
	PADS by Mentor Graphics	Licensed
	Proteus Design Suite by Lab Center Electronics	Licensed
	Pulsonix by Westdev Ltd	Licensed
	SIMBEOR by Simberian Inc.	Licensed
	TARGET 3001 by Ing.-Büero FRIEDRICH	Licensed
	Xpeditio Enterprise by Mentor Graphics	Licensed
	ZeniPCB by Stortini Mirko Bruno	Free
	Altium Designer by Altium	Licensed

some of the vendors is better in certain regions than in the others. So it is better to go for a vendor that has good reputation and history, introduces occasional updates and bug fixes, provides training courses and offers support in your region.

Integration. Flexibility or integration with other tools is an important factor to be considered. Make sure the software you select has functions to import and export files from and to formats supported by your target customers.

Interface. OrCAD, Eagle PCB and most open source software have easy-to-understand graphical user interface (GUI), which makes it easier for beginners and intermediates in the designing field. Some tools like Xpeditio use command-line interface, which can be a bit difficult for first-timers.

Free and open source tools

There are several free tools available online, individual software as well as design suites, that serve the purpose of circuit and embedded designing at different stages. Though these are quite popular among academia and hobbyists in electronics, the industry penetration of these software is very less; perhaps less than 5%.

One of the main reasons for the lesser popularity of free tools among professional designers is the limitation they provide in the number of pins, components, layers, multiple boards and so on. Hence these can be used only for low-end and mid-range schematic and PCB designing.

According to Rizwan Hirani of eInfoChips, compatibility with other layout tools is lesser for the free open source tools. The time taken for designing is usually more for such free tools. Performance, precision, training, support and documentation offered by licensed tools are also generally better than that available with the free tools. But in case of open source tools the users usually form a community to support and learn from each other. ●

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